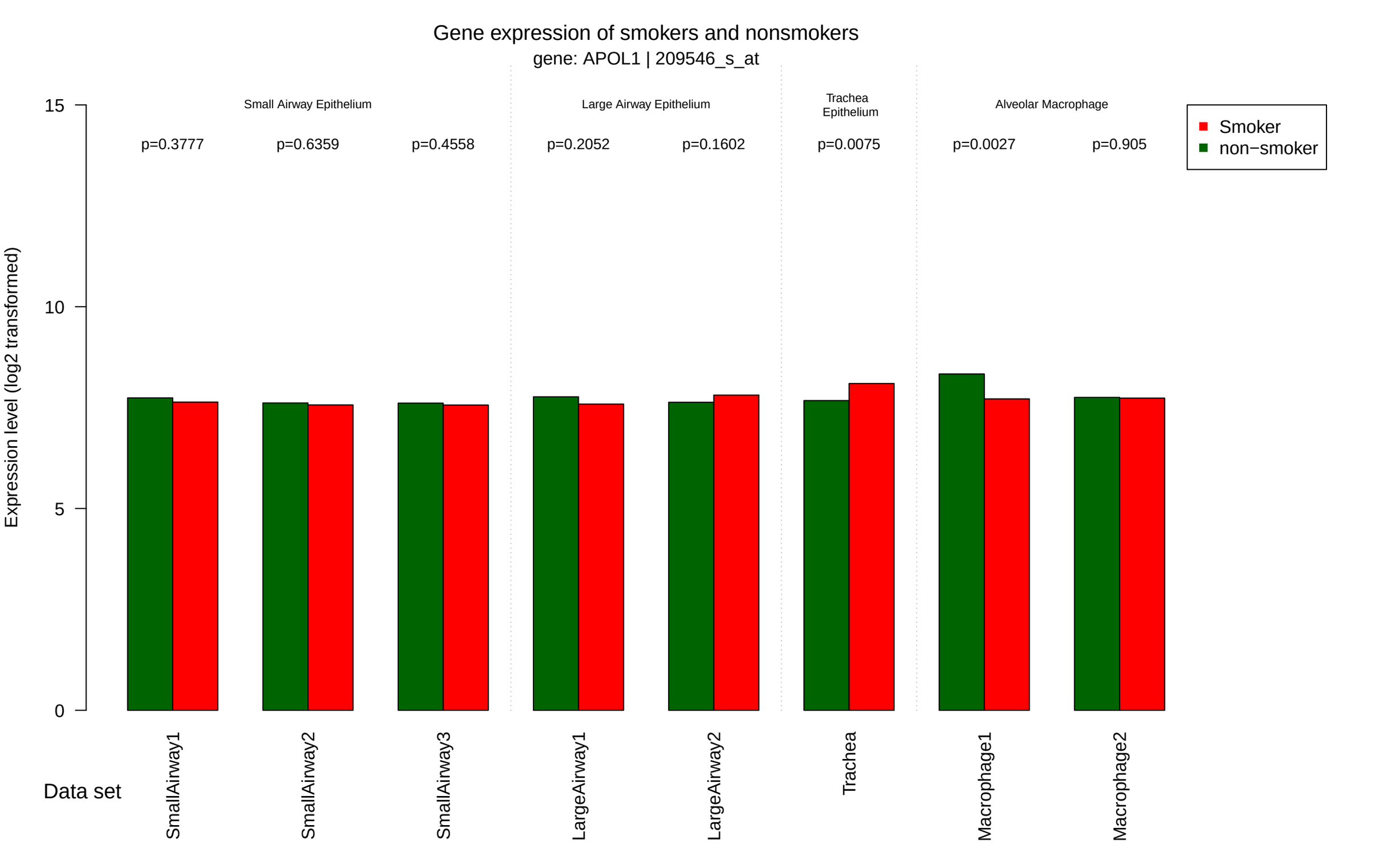
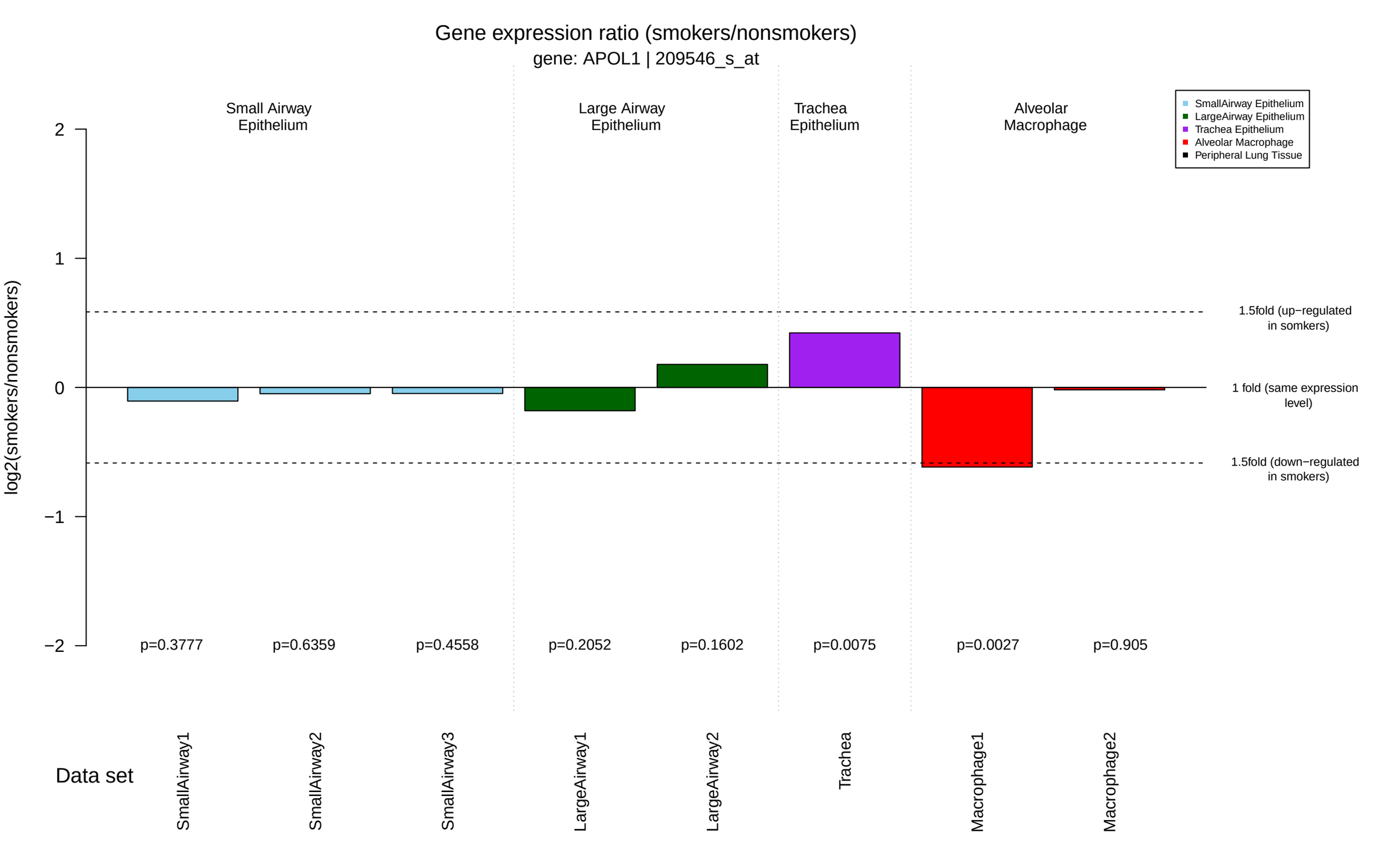
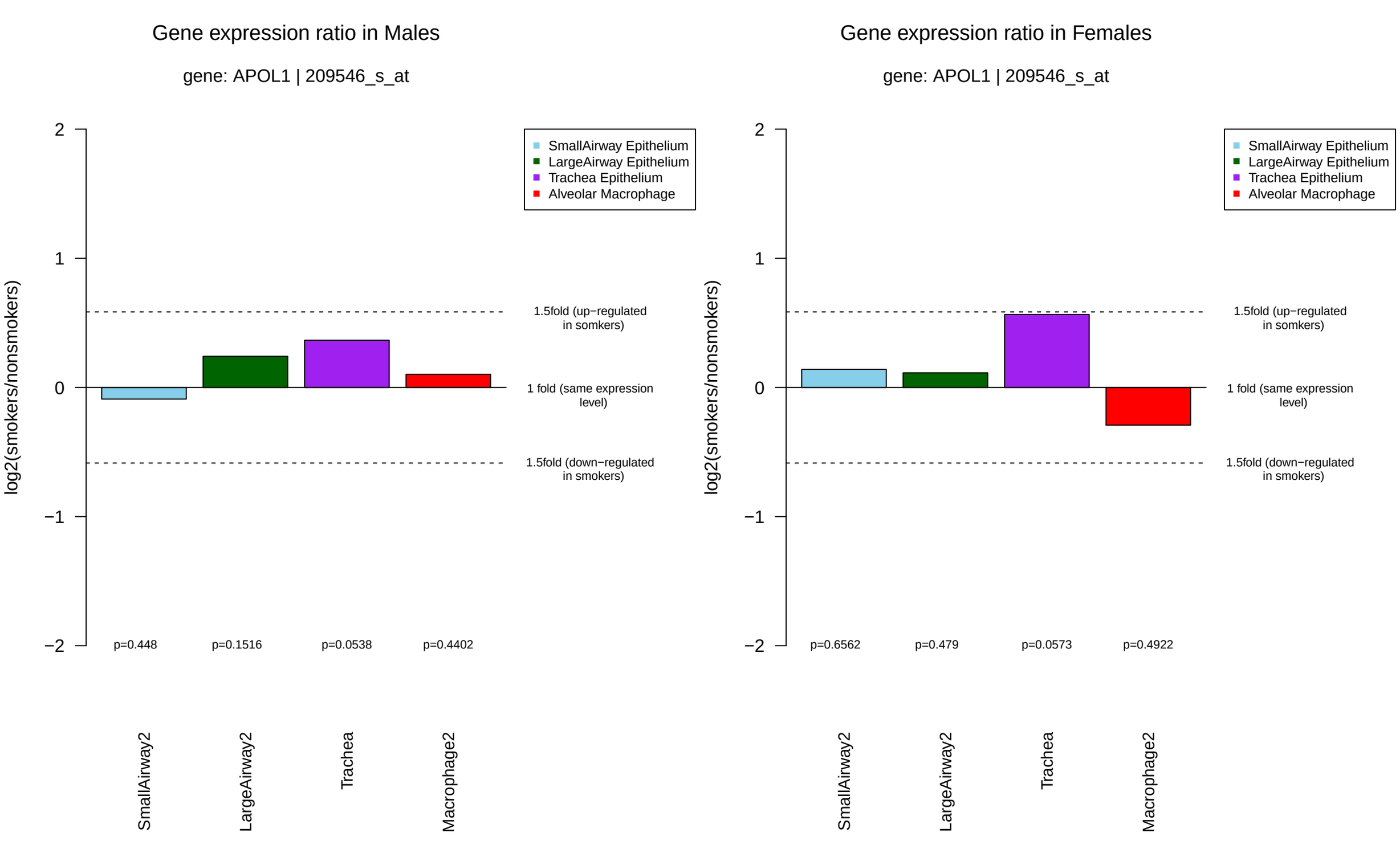
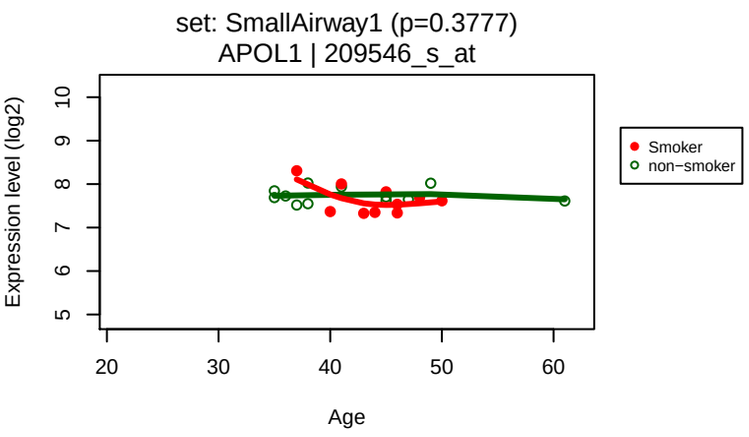
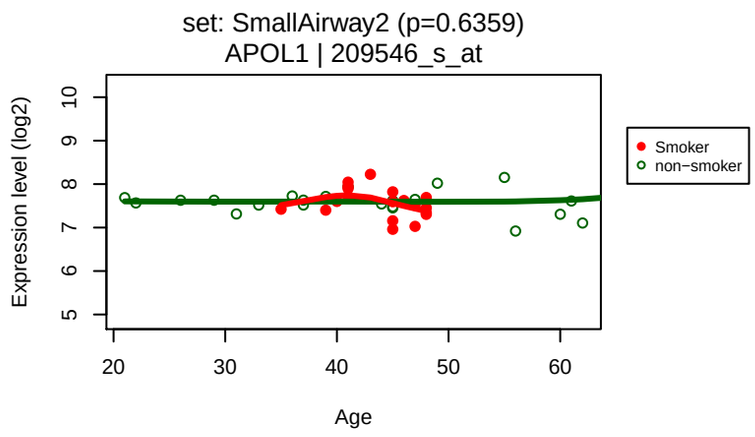
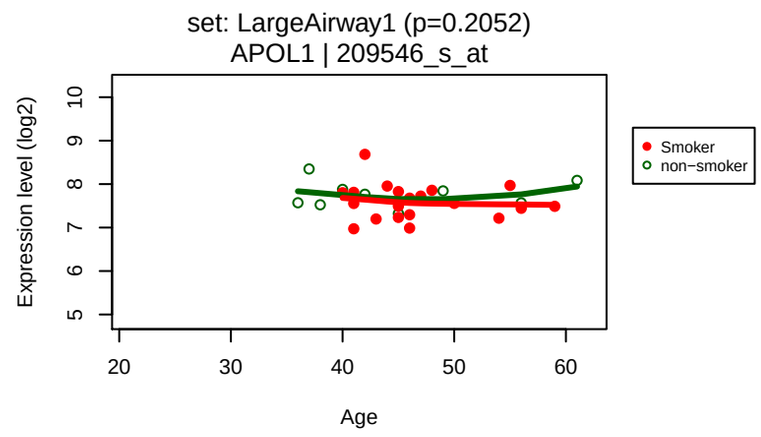

Gene expression of smokers and nonsmokers
gene: APOL1 | 209546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3777
p=0.6359
p=0.4558
p=0.2052
p=0.1602
p=0.0075
p=0.0027
p=0.905
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

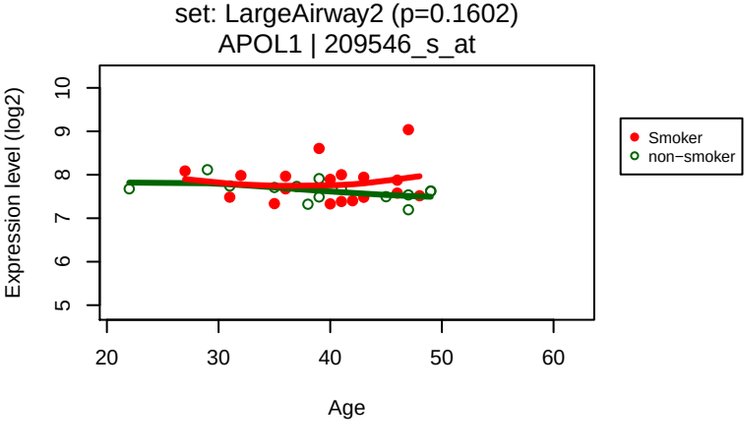

Gene expression ratio (smokers/nonsmokers)
gene: APOL1 | 209546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3777
p=0.6359
p=0.4558
p=0.2052
p=0.1602
p=0.0075
p=0.0027
p=0.905
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

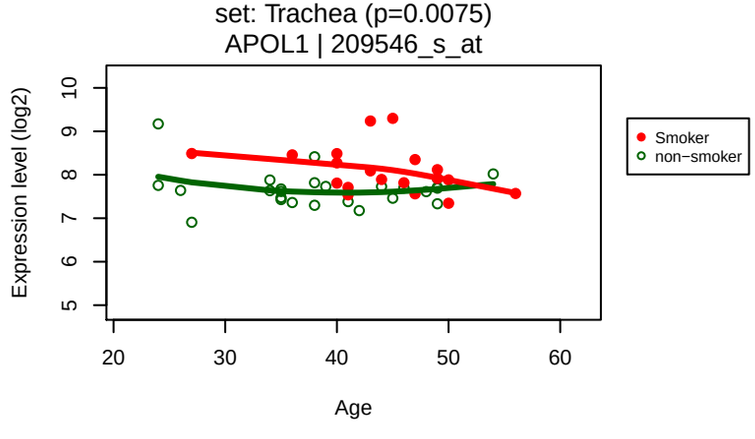

Gene expression ratio in Males
gene: APOL1 | 209546_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.448
p=0.1516
p=0.0538
p=0.4402
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOL1 | 209546_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6562
p=0.479
p=0.0573
p=0.4922
SmallAirway2
LargeAirway2
Trachea
Macrophage2

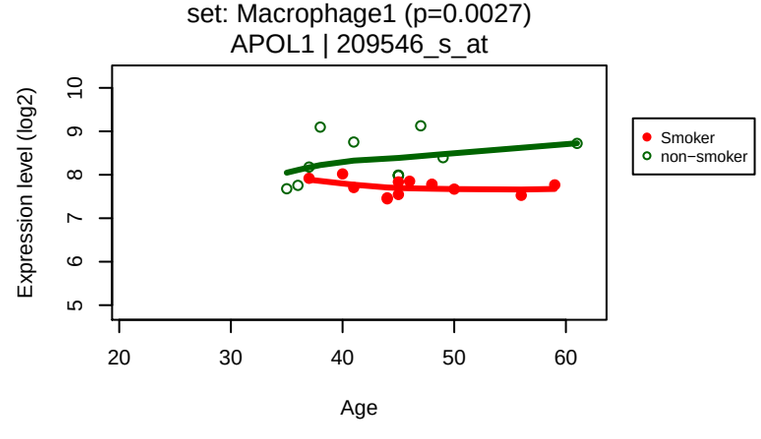

set: SmallAirway1 (p=0.3777)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker

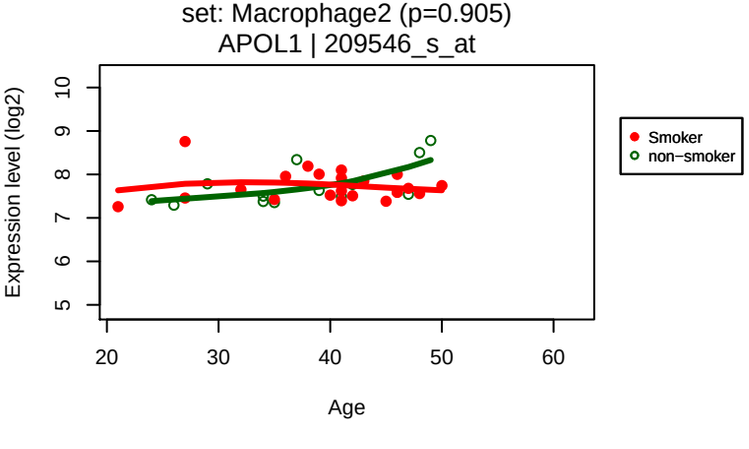

set: SmallAirway2 (p=0.6359)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker

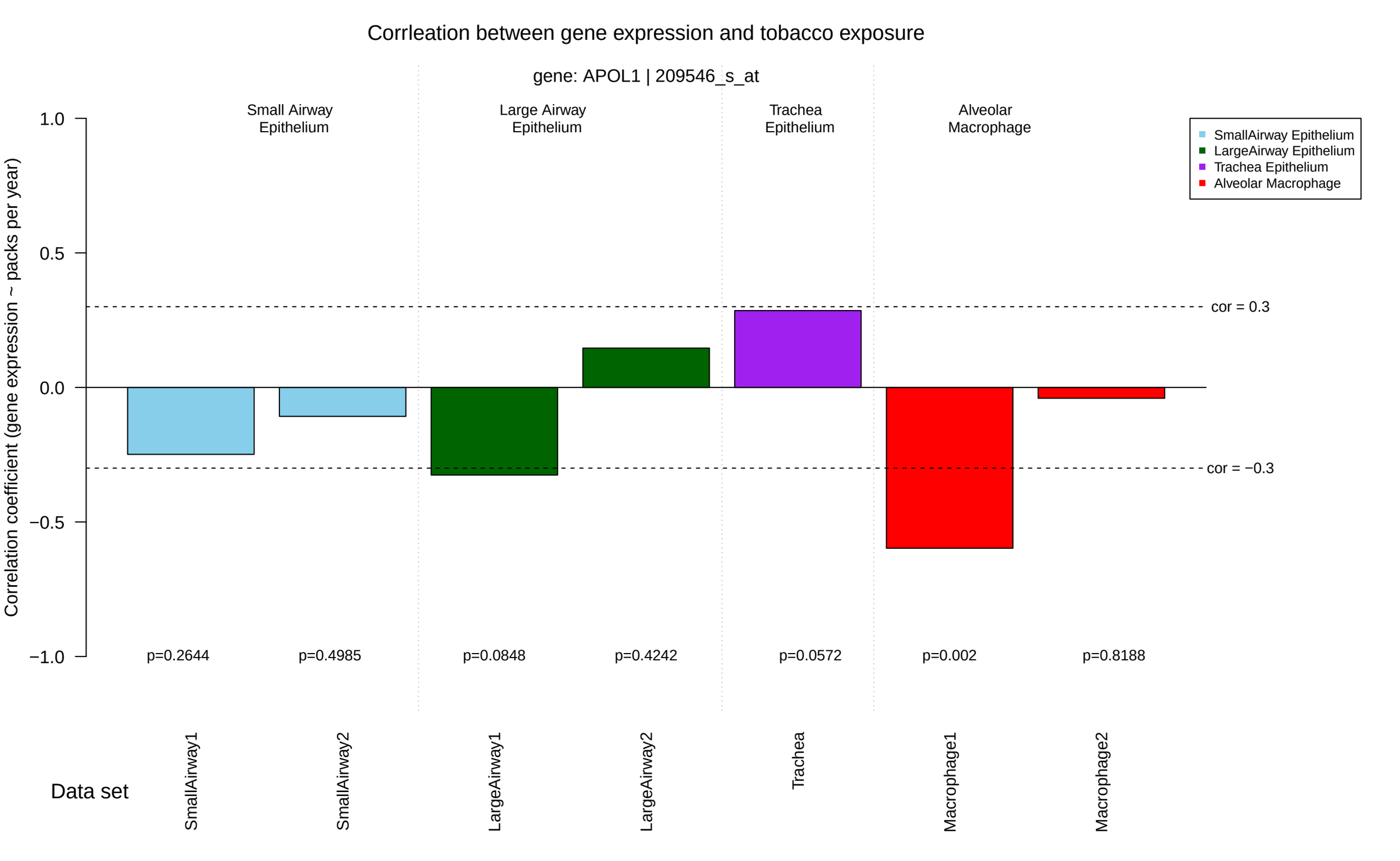

set: LargeAirway1 (p=0.2052)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1602)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.905)
APOL1 | 209546_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOL1 | 209546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2644
p=0.4985
p=0.0848
p=0.4242
p=0.0572
p=0.002
p=0.8188
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2